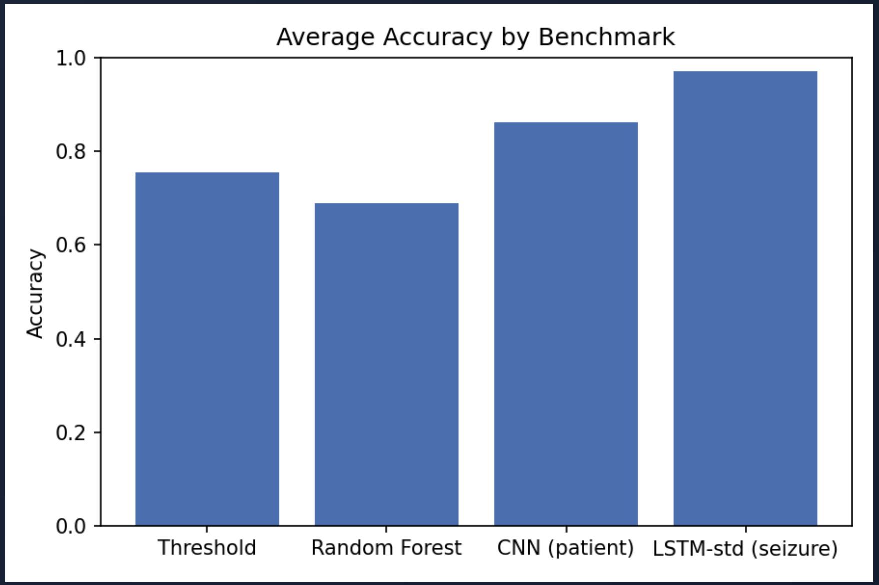
Deep Learning for EEG Seizure Detection

Project Report



Abstract

- **4 models**: threshold, random forest, CNN, LSTM
- **Patient-level** and **seizure-level** cross-validation
- 2-fold pilot: CNN leads at 79.7% acc
- LSTM results pending from GPU cluster

Contents

Introduction

Epileptic seizures = abnormal brain electrical activity detectable via EEG.

Goal: automatically detect seizures in EEG recordings.

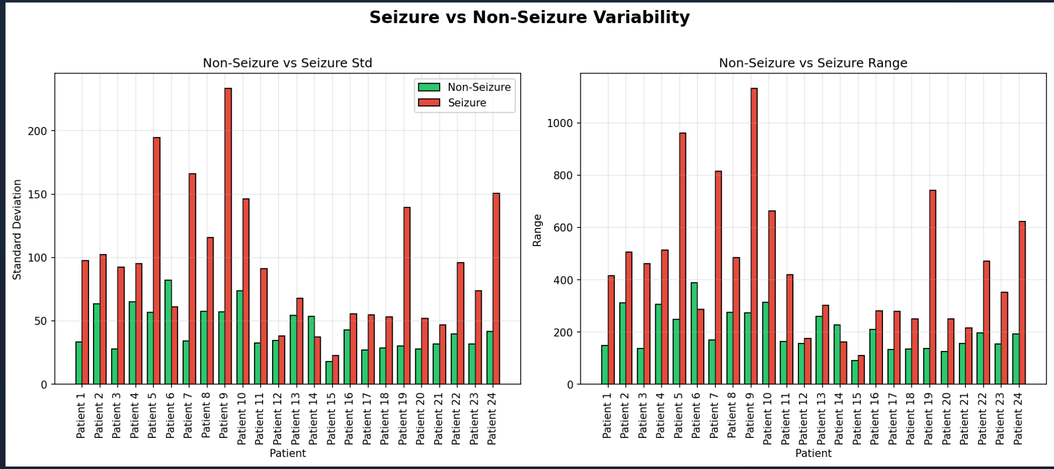
Approach: compare progressively richer models:

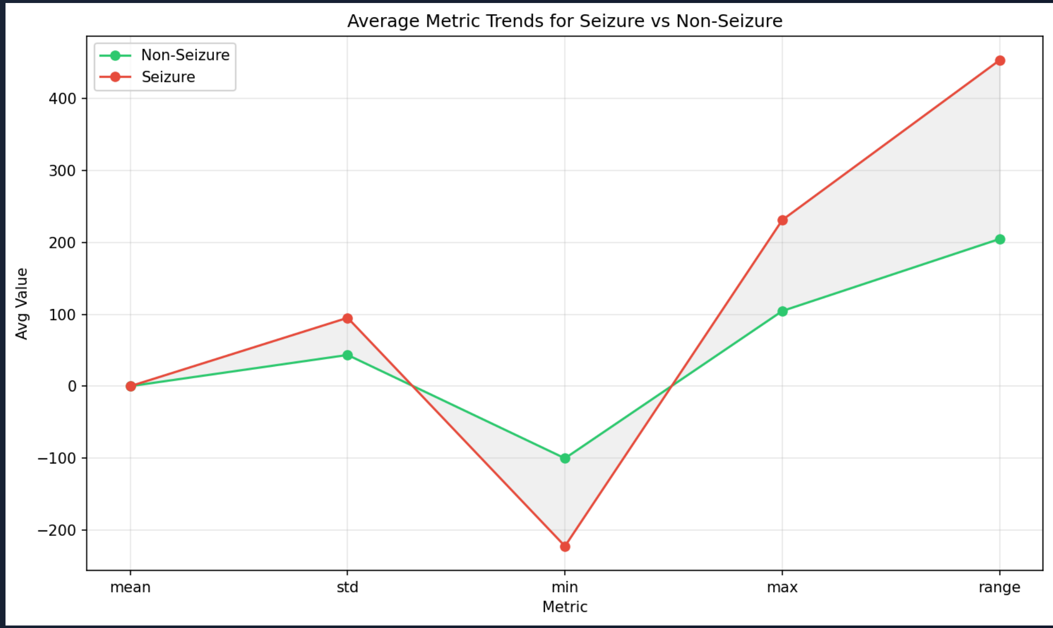
1. Threshold baseline (hand-crafted rules)
2. Random forest (hand-crafted features)
3. CNN (learned features, end-to-end)
4. LSTM (temporal episode modelling)

Dataset: CHB-MIT

- **Source**: CHB-MIT scalp EEG, 24 pediatric patients
- **Windows**: 1s segments → **571,905** total
- **Seizure**: **86,672** windows (15.2% — imbalanced dataset)
- **Shape**: each window = **21 channels × 128 timesteps**

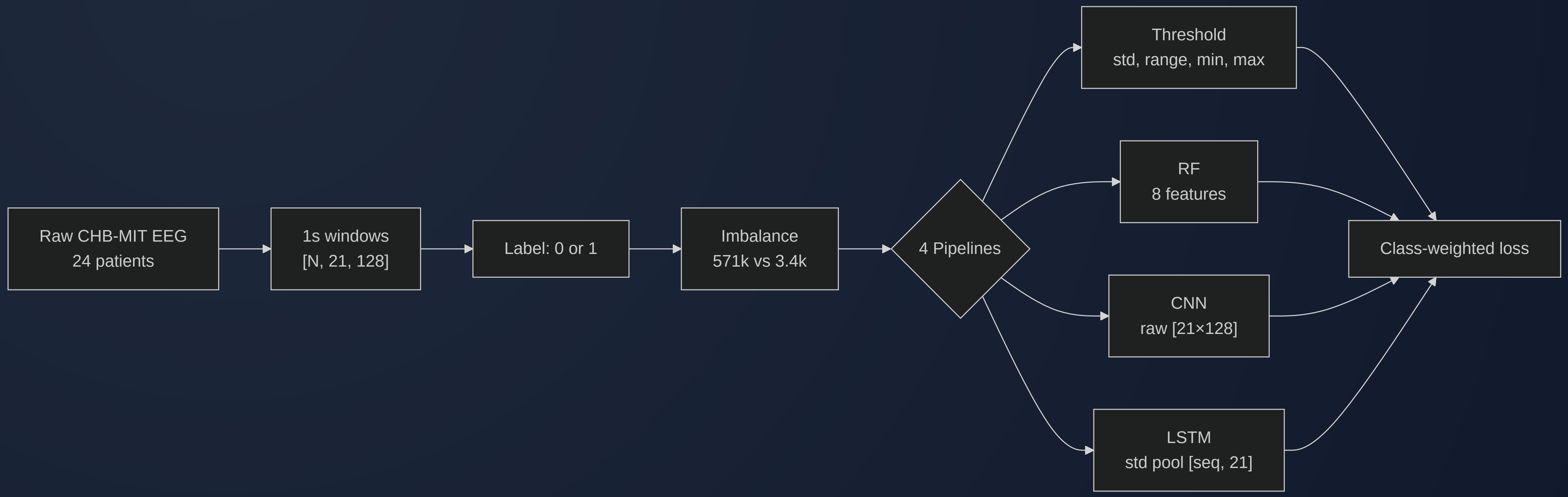
Dataset: Feature Trends





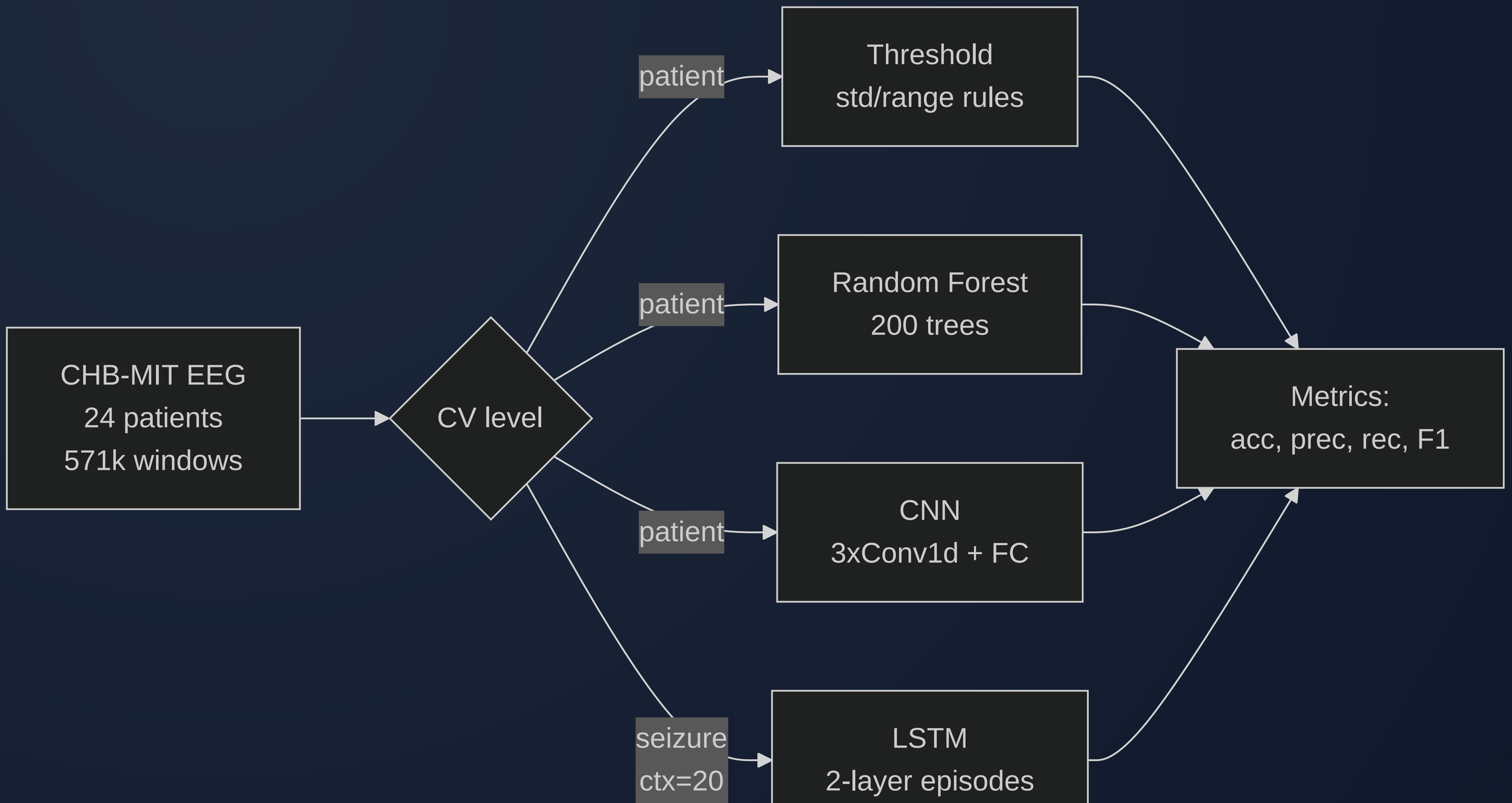
- **Std** and **range** separate seizure from non-seizure
- **Mean** ≈ 0 for both → risky feature alone

Preprocessing & Data Flow



- No per-window normalization (`normalize=False`)
- **Class weighting** helps models focus on the seizure minority

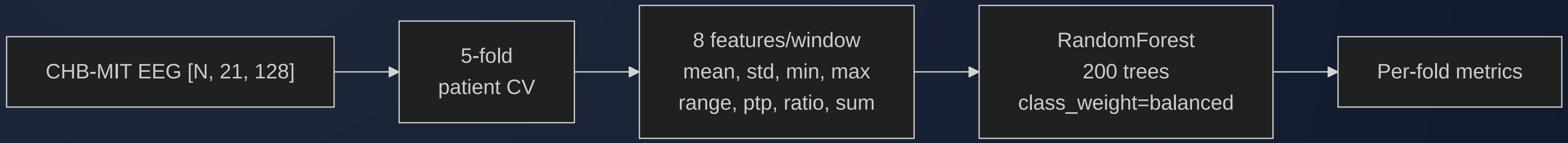
Overview: 4 Pipelines



Pipeline 1: Threshold Classifier

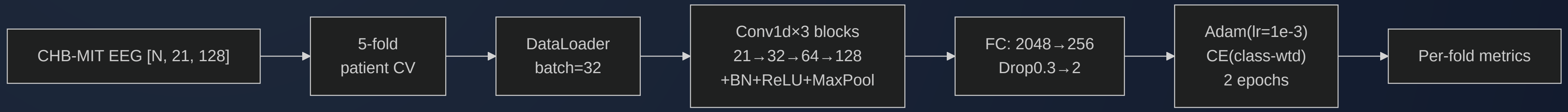
- Computes **std, range, min, max** per window
- Flags seizure if stat exceeds fold-specific threshold
- Thresholds = midpoint of seizure/non-seizure means

Pipeline 2: Random Forest



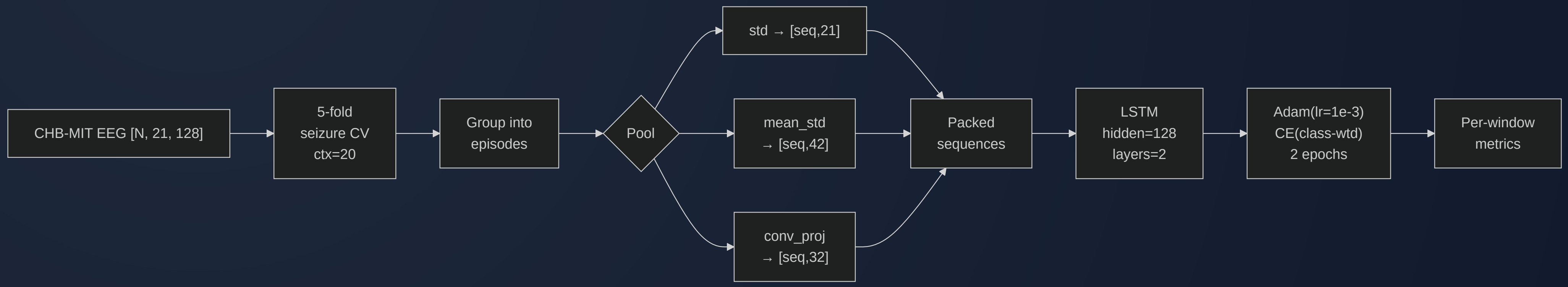
- **8 features**: mean, std, min, max, range, ptp, std/range ratio, range+std
- 200 trees, `class_weight='balanced'`
- Patient-level 5-fold CV

Pipeline 3: CNN



- **Input**: raw window [21, 128], no normalization
- 3× Conv1d → BN → ReLU → MaxPool
- FC: 2048→256→Dropout(0.3)→2
- Adam (lr=1e-3), class-weighted CE, 2 epochs

Pipeline 4: LSTM



- Groups contiguous same-label windows into **episodes**
- Per-window **pooling** reduces [21, 128] → feature vector
- 2-layer LSTM (hidden=128) + per-timestep classifier
- Seizure-level k-fold with context=20

Cross-Validation Strategies

Seizure-level

- Whole episode → one fold
- ±20 context windows in val

Window-level

- Window round-robin
- ±4 seizure neighbors dropped

Seizure-Level k-fold: Context

- Without context: val = **100% seizure** → no specificity measurable
- With `context=20`: ±20 non-seizure windows around each episode
- Val gets realistic ~40–60% non-seizure mix
- Train gets all remaining windows

LSTM: Pooling Options

EEG oscillates around 0 → **mean** collapses amplitude. **Std** preserves it.

Pool	Dim	Speed	Preserves
`std` (default)	21	Fast	Channel variance
`mean`	21	Fast	Near-zero — risky
`mean_std`	42	Fast	Center + spread
`conv_proj`	32	Medium	Learned features
`none`	2688	Very slow	Full waveform

LSTM: Architecture

```
Input:  [batch, seq_len, n_features]
              |
  LSTM(input=n_features, hidden=128, layers=2, dropout=0.3)
              |
  Linear(128 → 2) per-timestep
              |
Output: [batch, seq_len, 2]
```

- Packed sequences for variable-length episodes
- Class-weighted CE, `ignore_index=-1` for padding

LSTM: conv_proj Mode

Conv1d projection option replaces hand-crafted pooling:

```
[seq_len, 21, 128]
           |
  Conv1d(21 → 32, k=5, pad=2) → BN → ReLU
           |
  AdaptiveAvgPool1d(1)
           |
[seq_len, 32]  →  LSTM(input_size=32)
```

- Learns waveform features end-to-end
- Faster than `none` (2688-d), more expressive than `std` (21-d)

Hyperparameters

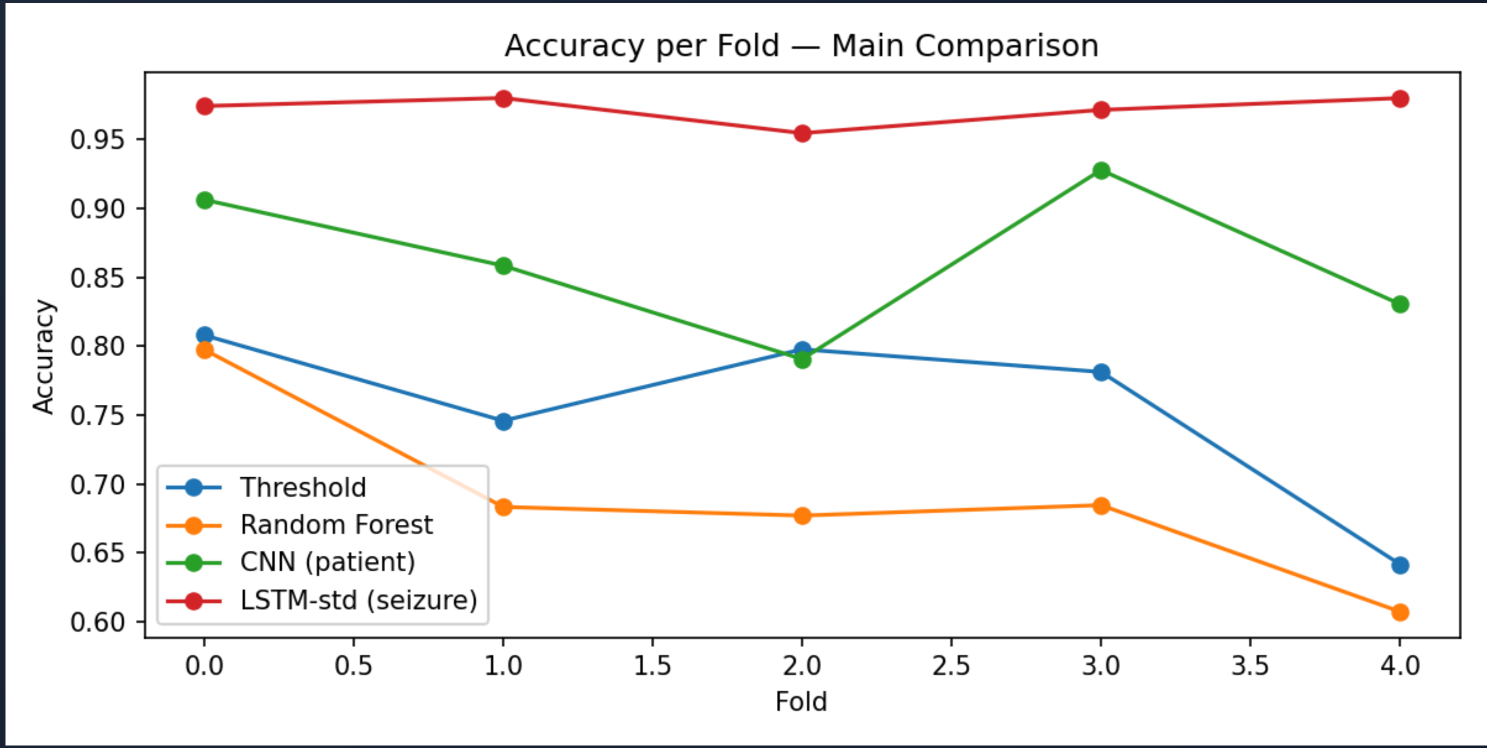
Parameter	Threshold	RF	CNN	LSTM
Trees/layers	—	200	[32,64,128]	2 LSTM
Hidden size	—	—	—	128
Pooling	—	—	—	std / conv_proj
Class weight	—	balanced	balanced	balanced
LR	—	—	1e-3	1e-3
Batch size	—	—	32	32
Epochs	—	—	2	2
Dropout	—	—	0.3	0.3
CV level	patient	patient	patient	seizure (ctx=20)

Experiment Grid

Script	Runs	~Time
`exp1_cnn_levels`	CNN × 3 levels	~1.5h
`exp2_lstm_levels_pool`	LSTM × 3 levels + 4 pools	~5-6h
`exp3_lstm_nsl`	LSTM × 4 NS lengths	~3.5h

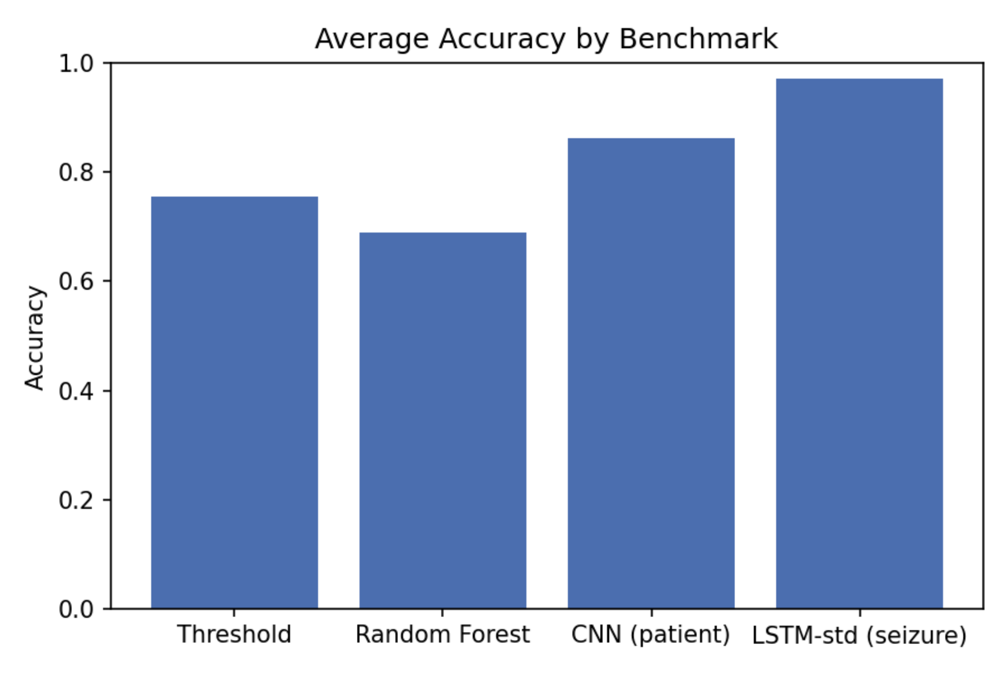
All write to `out/benchmarks/` with descriptive suffixes.

Results: Accuracy per Fold (Patient-Level)



High cross-patient variance across all models; CNN ranges 79–93%.

Results: Main Comparison (5-fold)



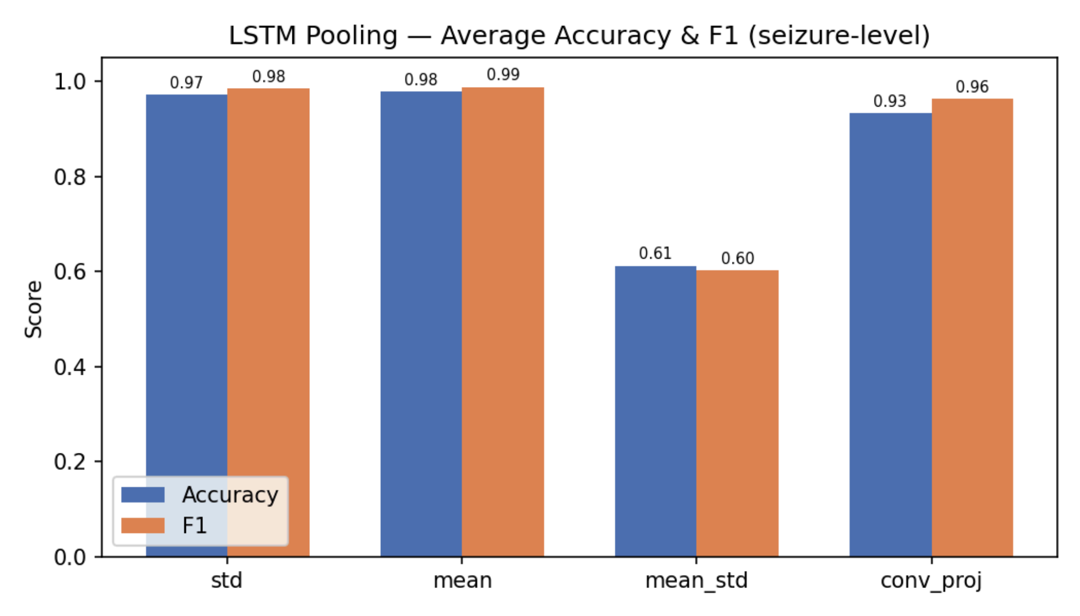
Model	CV	Acc	F1
Threshold	patient	75.5%	—
RF	patient	69.0%	—
CNN	patient	86.3%	0.554
CNN	seizure	80.1%	0.888
LSTM-std	seizure	**97.2%**	**0.985**

Results: Window-Level CV Degeneracy

CNN and LSTM under window-level CV: **97-100% accuracy but 0% F1**

- Round-robin assignment creates unreliable per-fold class ratios
- Models learn to predict "non-seizure" for everything → trivially high accuracy
- **Window-level CV is unsuitable** for this dataset

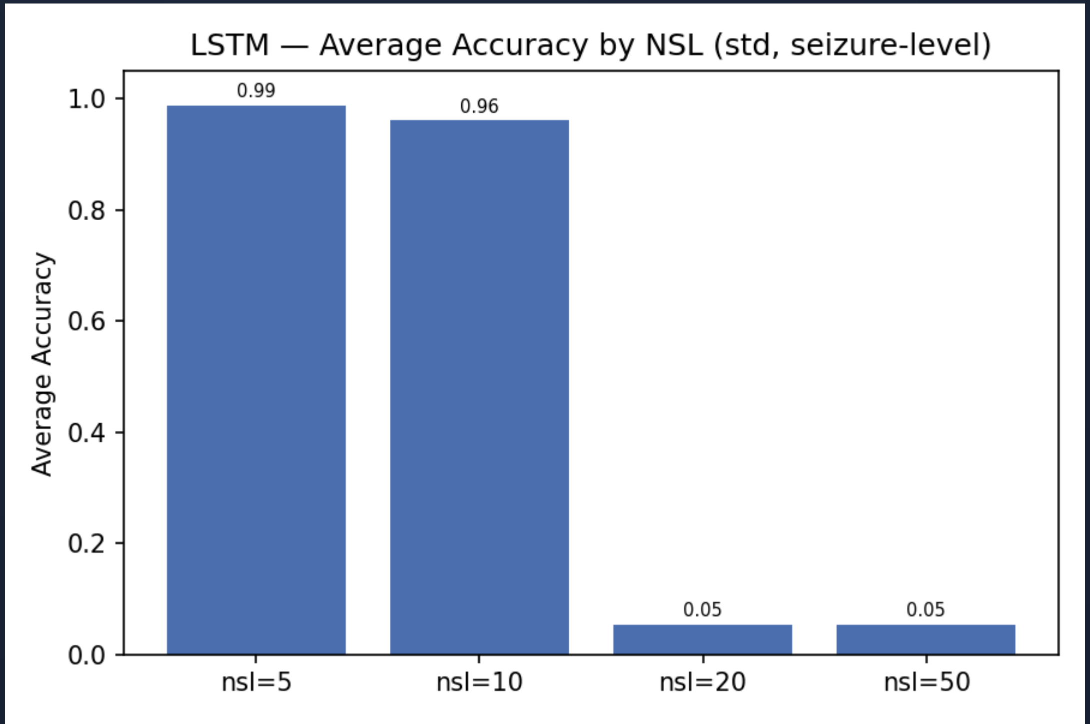
Results: LSTM Pooling (Seizure-Level)



Pooling	Avg Acc	Avg F1
std (21-d)	97.2%	0.985
mean (21-d)	**97.8%**	**0.988**
conv_proj (32-d)	93.3%	0.963
mean_std (42-d)	61.2%	0.601

> mean_std unstable (2/5 folds fail)

Results: Non-Seizure Length Sweep



NSL	Avg Acc	Avg F1
5	**98.8%**	**0.994**
10	96.1%	0.979
20	5.5%	0.000
50	5.5%	0.000

> NSL ≥ 20 → training collapse

Key Observations

- **LSTM (seizure-level)** dominates: 97.2% acc, 0.985 F1
- **CNN (patient-level)**: 86.3% acc but F1 only 0.554 — low recall on some patients
- **Window-level CV** produces degenerate results for all models
- **Threshold (75.5%)** outperforms RF (69.0%)
- **Mean & std pooling** both work well; mean_std is unstable
- **NSL ≥ 20** causes LSTM training collapse

Conclusion

- LSTM (seizure-level) achieves best results: **97.2% acc, 0.985 F1**
- CNN (patient-level): 86.3% acc, 0.554 F1 — high variance across patients
- Window-level CV is degenerate — poor class ratios per fold
- Mean pooling slightly outperforms std for LSTM
- **NSL ≤ 10** is essential; NSL ≥ 20 causes collapse
- **Future**: explore multichannel spatial info, longer temporal context

Thank You

Any Questions?